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Day 6

① SequentialSearch.java

array created int n = 1000;
for(int i=0; i<n; i++)
a[i] = i*2

0 2 4 6 8 ...

case 1: key = 0
Search(0,0)
i=0 a[0]=0 0==0 → found O(1)

case 2: key = 1998
i=0 → 0 != 1998
i=1 → 2 != 1998
i=2 → 4 != 1998
i=999 → 1998 == 1998 O(n) <- worst

case 3: key % = 7
0 != 7
2 != 7
4 != 7
return -1 O(n) <- worst

② Binary Search.java

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Array = [2, 5, 8, 12, 16, 23, 38]
key = 23

lo = 0
hi = 6
mid = (0+6) / 2 = 3
a[3] = 12

23 > 12 go right

lo = 4
hi = 6
mid = 5
a[5] = 23
23 == 23
return 5

Index = 5

O(log n)

③ Search Comparison

n = 1 000 000
queries = 2,000

for every key → start from index 0
↓
check each element
↓
stop when found

This is repeated 2000 times

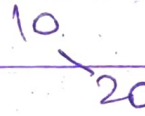
Unsorted array
↓
Arrays.sort()
↓
Sorted array

Linear search → O(n)
binary search → O(log n)
Sorting once → O(n log n)

④ AVL tree

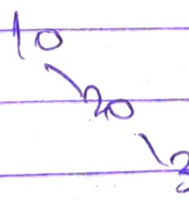
10 20 30 40 50 25

20 > 10



Balance 10 = 0 - 1 = -1

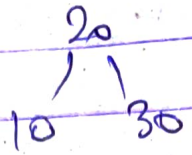
30 > 10 , 30 > 20



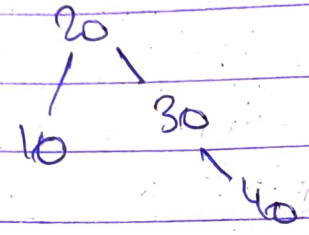
balance = -2

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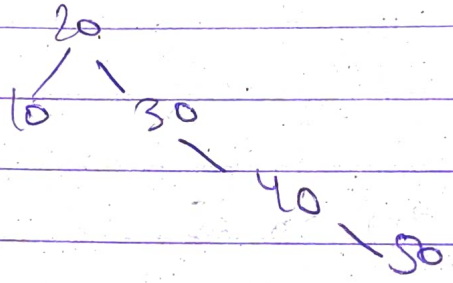
left rotate

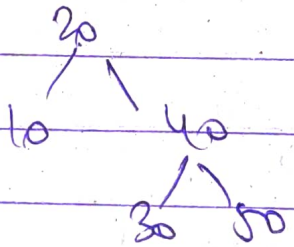


40 > 20 , 40 > 30

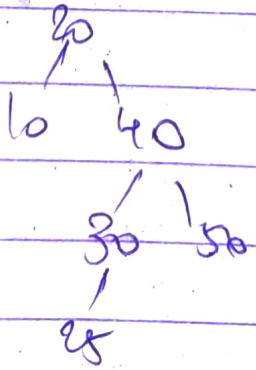


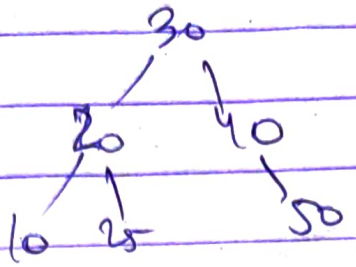
50 > 20 50 > 30 50 > 40



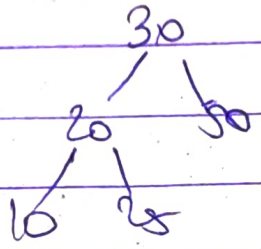


25 > 20 25 < 40 25 < 30



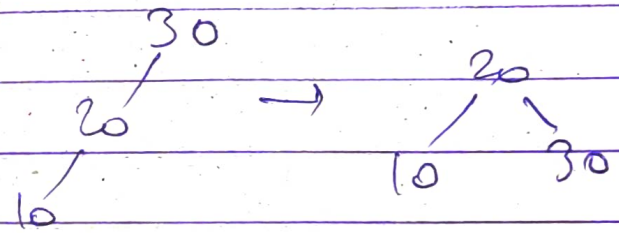


delete 40

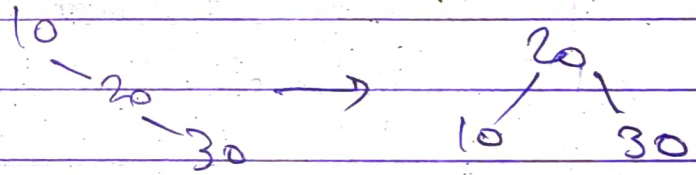


⑤ Four AVL cases

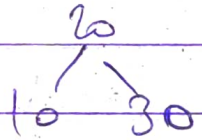
LL: 30 20 10



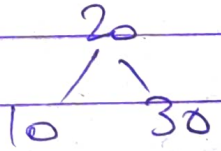
RR - 10 20 30



LR - 30 10 20

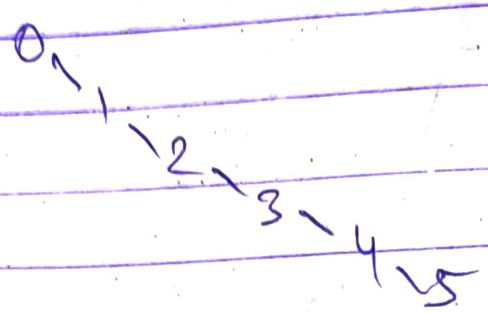


RL - 10 30 20

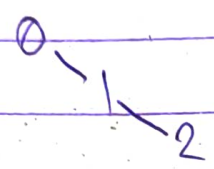


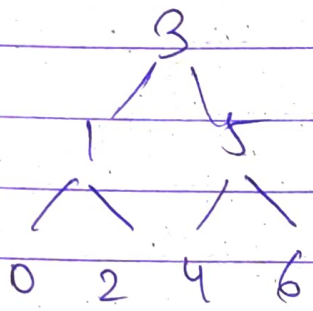
⑥ AVL vs BST

0 1 2 3 4 5 ...



AVL -





BST - o(n) can become
AVL - o(log n)

(9) JavaSearch Demo

Arrays.binarySearch ([2, 5, 8, 12 --], 23)
→ returns index 5

indexOf ("Meera")
→ checks names one by one
→ returns 2

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TreeMap.floorEntry(69)

keys = 0, 40, 55, 70, 85

largest key <= 69 = 55 → Grade C